

Education

University of Chicago

BA in Mathematics (Honors)

Chicago, IL
September 2010 - June 2014

University of California, Berkeley

Ph.D. in Statistics

Berkeley, CA
August 2014 - August 2018

Employment

Northwestern University, Department of Statistics and Data Science

Assistant Professor

Evanston, USA
September 2022 - present

- Assistant professor in the Department of Statistics and Data Science at Northwestern University.

Mayo Clinic

Visiting Research Scientist

Rochester, USA
September 2025 - present

- Visiting research scientist in Gastroenterology and Hepatology department.

Toyota Technological Institute at Chicago (TTIC)

Research Assistant Professor

Chicago, USA
September 2020 - August 2022

- Research Assistant Professor at the Toyota Technological Institute at Chicago.

Vector Institute

Postdoctoral Scholar

Toronto, Canada
August 2018 - July 2020

- Postdoctoral researcher at the Vector Institute at the University of Toronto

OpenAI

Research Scientist

San Francisco, USA
March 2016 - November 2017

- Full-time research scientist at OpenAI
- Advised by Ilya Sutskever

AI Expert at Y Combinator

Part-time Partner

San Francisco, USA
January 2017 - January 2018

- Advised Y Combinator on the latest trends in AI
- Advised over 50 startups on AI and ML problems

Young Scholars Program/SESAME

Counselor

Chicago, USA
June 2011 - May 2013

- Taught mathematics to Chicagoland high school students and CPS teachers
- Full-time during summers. Every Saturday and Sunday during school year
- Head counselor from June 2012 - May 2013

Conference Publications

Third-Person Imitation Learning

Bradly C. Stadie, Pieter Abbeel, Ilya Sutskever

In the proceedings of the International Conference on Learning Representations (ICLR), Toulon, France, April 2017

arXiv: <https://arxiv.org/pdf/1703.01703.pdf>

One-Shot Imitation Learning

Yan (Rocky) Duan, Marcin Andrychowicz, Bradly C. Stadie, Jonathan Ho, Jonas Schneider, Ilya Sutskever, Pieter Abbeel, Wojciech Zaremba

In Neural Information Processing Systems (NeurIPS), Long Beach, California, December 2017

arXiv: <https://arxiv.org/pdf/1703.07326.pdf>

Some Considerations on Learning to Explore via Meta-Reinforcement Learning

Bradly C. Stadie, Ge Yang, Rein Houthooft, Xi Chen, Yan Duan, Yuhuai Wu, Pieter Abbeel, Ilya Sutskever

In Neural Information Processing Systems (NeurIPS), Montreal, Canada, December 2018

arXiv: <https://arxiv.org/pdf/1803.01118.pdf>

Evolved Policy Gradients

Rein Houthooft, Richard Y. Chen, Phillip Isola, Bradly C. Stadie, Filip Wolski, Jonathan Ho, Pieter Abbeel

In Neural Information Processing Systems (NeurIPS) [Spotlight], Montreal, Canada, December 2018

arXiv: <https://arxiv.org/pdf/1802.04821.pdf>

One-Shot Pruning of Recurrent Neural Networks by Jacobian Spectrum Evaluation

Matthew Zhang, Bradly C. Stadie

In International Conference on Learning Representations (ICLR), Addis Ababa, Ethiopia, April 2020

arXiv: <https://arxiv.org/pdf/1912.00120.pdf>

Learning Intrinsic Rewards as a Bi-Level Optimization Problem

Lunjun Zhang, Bradly C. Stadie, Jimmy Ba

Conference on Uncertainty in Artificial Intelligence (UAI), July 2020. Available at: https://www.auai.org/uai2020/proceedings/66_main_paper.pdf

Maximum Entropy Gain Exploration for Long Horizon Multi-goal Reinforcement Learning

Harris Chan, Silviu Pitis, Stephen Zhao, Bradly C. Stadie, Jimmy Ba.

In International Conference on Machine Learning (ICML), July 2020. arXiv: <https://arxiv.org/abs/2007.02832>

World Model as a Graph: Learning Latent Landmarks for Planning

Lunjun Zhang, Ge Yang, Bradly C. Stadie

In International Conference on Machine Learning (ICML) [Long Presentation], July 2021. <https://arxiv.org/pdf/2011.12491.pdf>.

Invariance Through Latent Alignment

Takuma Yoneda, Ge Yang, Matthew Walter, Bradly C. Stadie

In Robotics: Science and Systems, July 2022. <https://invariance-through-latent-alignment.github.io/>

To the Noise and Back: Diffusion for Shared Autonomy

Takuma Yoneda, Luzhe Sun, Ge Yang, Bradly C. Stadie, Matthew Walter

In Robotics: Science and Systems, July 2023. Available at: <https://arxiv.org/pdf/2302.12244.pdf>

Cold Diffusion on the Replay Buffer: Learning to Plan from Known Good States

Z Wang, T Oba, T Yoneda, R Shen, M Walter, BC Stadie

In Conference on Robotic Learning, October 2023. Available at: <https://proceedings.mlr.press/v229/wang23e.html>

Wonderful Team: Zero-Shot Physical Task Planning with Visual LLMs

Zidan Wang, Rui Shen, and Bradly C. Stadie

In Transactions on Machine Learning Research 2025. Available at <https://arxiv.org/abs/2407.19094>

Expert of Experts Verification and Alignment (EVAL) Framework for Large Language Models Safety in Gastroenterology

Mauro Giuffrè, Kisung You, Ziteng Pang, Simone Kresevic, Sunny Chung, Ryan Chen, Youngmin Ko, Colleen Chan, Theo Saarinen, Milos Ajcevic, Lory S. Crocè, Guadalupe Garcia-Tsao, Ian Gralnek, Joseph J.Y. Sung, Alan Barkun, Loren Laine, Jasjeet Sekhon, Bradly Stadie, Dennis L. Shung

In Nature Digital Medicine. Available at <https://www.nature.com/articles/s41746-025-01589-z>

Of Mice and Machines: A Comparison of Learning Between Real World Mice and RL Agents
Shuo Han, German Espinosa, Junda Huang, Daniel A. Dombreck, Malcolm MacIver, Bradly C. Stadie
In ICML 2025. Available here: <https://arxiv.org/abs/2505.12204>

Thoughts and Lessons on Using Visual Foundation Models for Manipulation
Ryan Chen, Ziteng Pang, and Bradly C. Stadie
In Transactions on Machine Learning Research 2025 (TMLR). Available here: <https://openreview.net/forum?id=o6mnkDzVuc>

D2 Actor Critic: Diffusion Actor Meets Distributional Critic
Lunjun Zhang, Shuo Han, Hanrui Lyu, Bradly C. Stadie
In Transactions on Machine Learning Research 2025 (TMLR). Available here: <https://arxiv.org/abs/2510.03508>

LAMP: Extracting Local Decision Surfaces From Large Language Models
Ryan Chen, Youngmin Ko, Zeyu Zhang, Catherine Cho, Sunny Chung, Mauro Giuffrè, Dennis L. Shung, Bradly C. Stadie
AISTATS 2026, Spotlight Paper. Available at: <https://arxiv.org/abs/2505.11772>

Workshop Publications

Incentivizing Exploration in Reinforcement Learning with Deep Predictive Models
Bradly C. Stadie, Sergey Levine, Pieter Abbeel
In Neural Information Processing Systems (NeurIPS) Deep RL Workshop, Montreal, Canada, December 2015
arXiv: <https://arxiv.org/pdf/1507.00814.pdf>

Transfer Learning for Estimating Causal Effects Using Neural Networks.
Bradly C. Stadie, Soeren R. Kuenzel, Nikita Vemuri, Varsha Ramakrishnan, Jasjeet S. Sekhon, Pieter Abbeel
INFORMS Annual Meeting ML and causal inference workshop (2019)
arXiv: <https://arxiv.org/pdf/1808.07804.pdf>

Learning to Learn from Flawed, Failed, and Figurative Demonstrations
Ge Yang, Bradly C. Stadie, Roberto Calandra, Pieter Abbeel, Sergey Levine, Chelsea Finn
In Neural Information Processing Systems (NeurIPS) Deep RL workshop [Spotlight], Montreal, Canada, December 2018.
Available at: <https://github.com/bstadie/All-Bradly-Stadie-Papers/blob/master/leaf.pdf>

Understanding Hindsight Goal Relabeling from a Divergence Minimization Perspective
Lunjun Zhang, Bradly C. Stadie
NeurIPS 2022 Deep RL workshop. Available at: <https://arxiv.org/pdf/2209.13046>

Preprints

Evolutionary System Prompt Learning for Reinforcement Learning in LLMs
Lunjun Zhang, Ryan Chen, Bradly C. Stadie
Preprint. Available at: <https://arxiv.org/abs/2602.14697>

TEMPO: Temporal Enforcement via Mode-Separated Policy Optimization for Trustworthy LLM Backtesting
Zeyu Zhang, Bradly C. Stadie
Submitted to NeurIPS 2026. Available at: <https://arxiv.org/abs/2605.18843>

TraceDx: Criticality-Weighted Atomic Facts as a Training Signal for Sequential Clinical Diagnosis
Jiayi Li, Sunny Chung, Don Chamil Codipilly, George Marek, Varan Peranathan, Dennis L. Shung, Bradly C. Stadie
Submitted to NeurIPS 2026. Available at: <https://bstadie.github.io/files/simulator.pdf>

Schema Guidance as Managed Context for Enhanced Agentic SQL Generation in Clinical Databases

Dennis Shung, Bradly C. Stadie

Submitted to NeurIPS 2026.

Available at: <https://bstadie.github.io/files/maestro.pdf>

All Leaks Count, Some Count More: Interpretable Temporal Contamination Detection and Mitigation in LLM Back-testing

Zeyu Zhang, Ryan Chen, Bradly C. Stadie

Submitted to EMNLP 2026.

Available at: <https://arxiv.org/abs/2602.17234>

Disentangling Writing-Style Bias in LLMs: Complexity Narrows Substance, Errors Warm Tones

Youngmin Ko, Bradly C. Stadie

Submitted to EMNLP 2026.

Available at: <https://bstadie.github.io/files/typos.pdf>

Technical Reports

Simulating the Stochastic Dynamics and Cascade Failure of Power Networks

Charles Matthews, Bradly C. Stadie, Jonathan Weare, Mihai Anitescu, Christopher Demarco

Technical Report. Available at: <https://arxiv.org/pdf/1806.02420.pdf>

One-Demonstration Imitation Learning

Bradly C. Stadie, Siyan Zhao, Qiqi Xu, Bonnie Li, Lunjun Zhang

Technical Report. Available at: <https://github.com/bstadie/All-Bradly-Stadie-Papers/blob/master/one-demo.pdf>

AIA Forecaster: Technical Report

Rohan Alur, Bradly C. Stadie, Daniel Kang, Ryan Chen, Matt McManus, Michael Rickert, Tyler Lee, Michael Federici, Richard Zhu, Dennis Fogerty, Hayley Williamson, Nina Lozinski, Aaron Linsky, Jasjeet S. Sekhon

Technical Report. Available at: <https://arxiv.org/abs/2511.07678>

Selected Talks

- **CFMG Workshop Northwestern University (January 2026)** Embodiment Guided VLA
- **Digestive Disease Week (May 2025)** Automated Prompt Optimization Strategy Improves Large Language Model Diagnostic Accuracy for Complex Clinical Cases in Gastroenterology and Hepatology
- **NITMB External Advisory Board Meeting (April 2024)** What is the Place for Planning?
- **Midwest Robotics Workshop (April 2024)** Re-examining Actor Critic Methods in the 2020s.
- **Workshop: NITMB Kickoff Workshop (November 2023)** Theme Talk: Machine Learning.
- **Northwestern University (February 2023)** Making Sense of the Past with Graph Based Planning and Cold Diffusion.
- **Workshop: Foundation Models for Decision Making Hindsight Divergence Minimization.** December 2022.
- **Workshop: Deep Reinforcement Learning Workshop.** Hindsight Divergence Minimization. December 2022.
- **Toyota Technological Institute Japan Conference on Symbolic Neural Learning, Nagoya Japan, Invited Speaker (July 2022):** Making Sense of the Past with Hindsight Divergence Minimization
- **NSF EPSCoR Workshop: Artificial Intelligence With no Boundary Thinking, Invited Speaker (April 2022):** Making Sense of the Past with Hindsight Divergence Minimization
- **MSRI Hot Topics Workshop, Invited Speaker (March 2022):** Sim2Real Transfer in Robotics: Thoughts on Model Pruning and Robust Visual Transfer
- **Northwestern University (January 2022):** Towards Strong Generalization in Reinforcement Learning.
- **US Naval Academy (USNA) (January 2022):** Towards Strong Generalization in Reinforcement Learning.
- **UC Berkeley (October 2020):** Weakly Supervised Reinforcement Learning

- **Toyota Technological Institute, Chicago (March 2020):** Weakly Supervised Reinforcement Learning
- **Northwestern University invited speaker (October 2019):** Learning reward functions.
- **INFORMS panel on ML in causal inference (October 2019):** Transfer learning causal effects.
- **University of Illinois, Urbana Champaign (October 2019):** Learning reward functions.
- **Toyota Technological Institute, Chicago (May 2019):** Learning from imperfect data.
- **University of Wisconsin, Madison (April 2019):** Learning from imperfect data.
- **Purdue University (April 2019):** Learning from imperfect data.
- **University of Montreal (February 2018):** The relationship between sampling and meta learning.
- **University of Toronto (February 2018):** The relationship between sampling and meta learning.
- **Harvard University 2017 Big Data Conference, Invited Speaker (August 2017):** Frontiers of meta learning.
- **OpenAI (March 2016):** Exploration in deep reinforcement learning.

Selected Teaching

Courses Taught (Northwestern)

- **Statistics 436: Reinforcement Learning.**
- **Statistics 415: Introduction to Machine Learning.**
- **Statistics 360: Introduction to Generative AI.**
- **Statistics 320: Statistical Theory and Methods.**

Past Teaching

- **Statistics 133: Concepts in Computing with Data.** Graduate Student Instructor. *Jan - May 2015*
- **Statistics 088: Introduction to Data Science.** Graduate Student Instructor. *Jan - May 2016*
- **TTIC 31170: Planning, Learning, and Estimation for Robotics and Artificial Intelligence.** Co-Instructor. *March 2021 - June 2021*
- **TTI, Nagoya Japan - Introduction to Artificial Intelligence and Machine learning.** Selected Lectures on Deep Learning. *July 2022*